

Whats App

Functional requirements:

user should be able to send image/videos/files based on his preference,
user should be able to search for some chat based on his preference,
user should be able to chat actively based on his preference,

Non-functional requirements:

data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine i.e 99.99% uptime i.e Availability>>Consistency)

Core Entities:

user ,
media,
filemetadata

API :

POST:/upload

```
{  
  userID,  
  filemetadata  
}
```

GET:/search/{fileID}

HLD:

